

# ARCHAEOLOGY

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity and pick up an archaeologist's shovel. The principle of stratigraphy says that deeper layers are usually older, and carbon-14 dating can date organic remains such as bone and wood. Göbekli Tepe in Turkey surprised researchers because its stone temples are older than farming. Cuneiform writing arose in Mesopotamia, and the Ishtar Gate, with blue bricks, dragons and lions, led into Babylon. At Ugarit a tablet was found with the oldest written melody – a hymn to the goddess Nikkal. The Indus Valley script is still undeciphered, and at Mohenjo-daro a small bronze figurine called the 'Dancing Girl' was found. Linear B was deciphered in 1952 by Michael Ventris – it turned out to be an early form of Greek. The palace of Knossos on Crete was excavated in 1900–1905 by Arthur Evans. The pyramids of Giza were built as tombs for pharaohs, and Egyptians moved huge blocks on sleds, pouring water onto the sand in front of them. In 1922 Howard Carter discovered Tutankhamun's tomb, and in 1939 the BBC broadcast the pharaoh's instruments – trumpets were played. In the 'Book of the Dead', the heart of the deceased was weighed against the feather of the goddess Maat. Ötzi, the Iceman of the Alps, carried an ax with a copper blade. The smaller 'bluestones' of Stonehenge were brought from Wales, about 250 km away. The Terracotta Army guards the tomb of the first emperor of China, and in 79 AD Pompeii was buried under a thick layer of ash and pumice. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 Which instruments from Tutankhamun's tomb were played on a BBC radio broadcast in 1939?
  - a) Drums
  - b) Trumpets
  - c) Harps
  - d) Flutes
2. 👁 The Terracotta Army, made up of thousands of clay soldiers, guards the tomb of...
  - a) the first emperor of China
  - b) King David
  - c) Alexander the Great
  - d) the pharaoh Khufu
3. 📖 Where did cuneiform writing originate?
  - a) In China
  - b) In Greece
  - c) In Mesopotamia
  - d) In Mesoamerica
4. ⚖ Using radiocarbon (C-14) dating, archaeologists can date...
  - a) bronze coins
  - b) organic remains such as bone and wood
  - c) gold ornaments
  - d) stone statues
5. 🗡 The smaller "bluestones" at Stonehenge were transported from...
  - a) the Alps via France
  - b) the neighboring valley
  - c) Wales (about 250 km away)
  - d) Egypt across the sea
6. 🧑 Who discovered Tutankhamun's intact tomb in the Valley of the Kings in 1922?
  - a) Howard Carter
  - b) Arthur Evans
  - c) Heinrich Schliemann
  - d) Jean-François Champollion
7. 💖 In the Egyptian Book of the Dead, the heart of the deceased was weighed against...
  - a) a gold coin
  - b) a stone from the Nile
  - c) the pharaoh's heart
  - d) the feather of the goddess Maat
8. 🍃 Ötzi the Iceman (c. 3300 BC), found in the Alps, was carrying, among other things,...
  - a) an iron sword
  - b) coins from Rome
  - c) an ax with a copper blade
  - d) a magnetic compass
9. ☆ The pyramids at Giza were built as...
  - a) star observatories
  - b) tombs for pharaohs
  - c) granaries
  - d) palaces for queens

# PHILOSOPHY

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, when philosophy was born – in Greek the word means love of wisdom. Thales of Miletus believed that the first principle of everything is water, and Anaximander claimed that the Earth floats freely in space. The Pythagoreans discovered that pleasant harmonies come from simple number ratios, such as 2:1 and 3:2. In Zeno of Elea's paradox, swift Achilles can never catch a tortoise. On the Temple of Apollo at Delphi stood the words: 'Know thyself'. Socrates talked with Athenians in the agora, the marketplace, and said: 'I know that I know nothing!'. In his allegory of the cave, Plato compared people to prisoners who see only the shadows of things; he held that the soul is immortal and that music in education shapes character and the soul. His 'Symposium' is a conversation about love (eros). Aristotle's students, the Peripatetics, held discussions while walking with their master. Aristotle created the syllogism: if every man is mortal and Socrates is a man, then Socrates is mortal. The Cynic Diogenes reportedly lived in a barrel, a large clay jar. Epicurus taught that we need not fear death, because when death is here, we no longer are. The Stoics took their name from the Stoa Poikile, the painted porch; Marcus Aurelius belonged to Stoicism and wrote his 'Meditations' as notes to himself. In the East, Confucius taught respect, duty and good manners, and Laozi, legend says, rode off to the west on an ox. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The Pythagoreans discovered that pleasing musical harmonies result from...
  - a) the time of day
  - b) simple number ratios, such as 2:1 and 3:2
  - c) the color of the strings
  - d) the size of the hall
2. 👁 According to tradition, the Cynic philosopher Diogenes of Sinope lived in...
  - a) the king's palace
  - b) a mountain cave
  - c) a barrel (a large clay jar)
  - d) the temple of Apollo
3. 🗨 The word "philosophy" comes from Greek and means...
  - a) knowledge of the gods
  - b) love of wisdom
  - c) art of speaking
  - d) study of the stars
4. 🏃 In Zeno of Elea's paradox, the swift runner Achilles can never catch up with...
  - a) a horse
  - b) the wind
  - c) a tortoise
  - d) a hare
5. 🎓 Aristotle's school was called Peripatetic because his students...
  - a) studied on a ship
  - b) sat in silence
  - c) wrestled before lessons
  - d) held discussions while walking
6. 🗨 Plato's dialogue the Symposium is a conversation among friends at a banquet about...
  - a) shipbuilding
  - b) love (eros)
  - c) the war with Persia
  - d) maritime law
7. ❤ Which temple bore the maxim "Know thyself"?
  - a) The Temple of Apollo at Delphi
  - b) The Temple of Artemis at Ephesus
  - c) The Parthenon in Athens
  - d) The Pantheon in Rome
8. 🌿 Thales of Miletus believed that the first principle (arche) of everything was...
  - a) water
  - b) air
  - c) fire
  - d) number
9. ⭐ In the Allegory of the Cave, Plato compared people to...
  - a) prisoners who see only the shadows of things
  - b) sailors on a stormy sea
  - c) shepherds leading their flocks
  - d) builders of a tower

1. b 2. c 3. b 4. c 5. d 6. b 7. a 8. a 9. a

A N S W E R S

# HISTORY



## 4. Antiquity

5000 BC – 500 AD

Let's travel back to antiquity, to the age of great battles and rulers. In 490 BC, after the Battle of Marathon, legend says a messenger ran with news of victory to Athens. In 480 BC Leonidas and 300 Spartans defended the pass of Thermopylae, and the naval battle of Salamis was a victory of the Greeks over the Persians. Spartans marched into battle in step to the sound of the aulos (a pipe). The teacher of young Alexander the Great was Aristotle. In 218 BC Hannibal crossed the Alps with war elephants, breaking rocks by heating them with fire and pouring vinegar on them. Hatshepsut, a female pharaoh, was shown on statues wearing a pharaoh's beard, and Cleopatra VII was the last ruler of the Ptolemaic dynasty. According to the Bible, the walls of Jericho fell when trumpets made of rams' horns were blown. Julius Caesar, crossing the Rubicon, said 'Alea iacta est' – the die is cast; after a swift victory he wrote 'Veni, vidi, vici' – I came, I saw, I conquered. He was killed on the Ides of March, the 15th of March, 44 BC. The poet Ovid wrote his 'Tristia' in exile by the Black Sea. Hadrian's Wall cut across northern Britain. The Edict of Milan in 313 granted Christians freedom of worship. Tradition says Rome was founded in 753 BC, and the Western Empire fell in 476, when Odoacer deposed Romulus Augustulus – so 1228 full years passed, since there was no year zero. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🏹 According to the Bible, the walls of Jericho fell when the Israelites blew...
  - a) bronze bells
  - b) shepherds' flutes
  - c) war drums
  - d) trumpets made of rams' horns
2. 👁 Hadrian's Wall, built around AD 122, ran across...
  - a) northern Britain
  - b) the Alps
  - c) the Crimean Peninsula
  - d) the Nile Valley
3. 🗞 After a quick victory (47 BC), Julius Caesar wrote "Veni, vidi, vici," which means...
  - a) The die is cast
  - b) I came, I saw, I conquered
  - c) Divide and rule
  - d) Bread and circuses
4. ➡ According to tradition, Rome was founded in 753 BC, and the Western Roman Empire fell in AD 476. How many full years passed (remember: there was no year 0)?
 

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5. 🚩 According to legend, after the Battle of Marathon (490 BC), a messenger ran with news of the victory to...
  - a) Sparta
  - b) Rome
  - c) Athens
  - d) Troy
6. 🧠 The teacher of the young Alexander the Great was...
  - a) Socrates
  - b) Pythagoras
  - c) Aristotle
  - d) Plato
7. 💜 The Roman poet Ovid wrote his sorrowful Tristia while...
  - a) in exile by the Black Sea
  - b) on an expedition to Egypt
  - c) at Emperor Augustus's court
  - d) on vacation in Sicily
8. 🍀 In 218 BC, Hannibal crossed the Alps with an army and...
  - a) Arabian horses
  - b) lions
  - c) war elephants
  - d) camels
9. ☆ Emperor Constantine's Edict of Milan (313)...
  - a) moved the capital to Athens
  - b) banned all religions
  - c) divided the empire
  - d) granted Christians freedom of worship

# LINGUISTICS

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, to the birth of alphabets and words. The Greek alphabet was based on that of the Phoenicians; Phoenician was read from right to left, and Greek from left to right. The classical Greek alphabet has 24 letters, and the word 'alphabet' joins alpha and beta. Chinese writing is logographic: a sign stands for a word or idea. Pharaohs' names in hieroglyphs were enclosed in an oval frame called a cartouche. The Egyptians simply called the Nile 'Iteru' – the great river. Greek gave us many words: 'democracy' is rule by the people, 'hippopotamus' is river horse, and 'music' meant the art of the Muses. The 'Iliad' and the 'Odyssey' were recited from memory by wandering rhapsodes. Roman pupils wrote with a stylus on wax-covered tablets, and long scrolls were unrolled with one hand and rolled up with the other. Cicero left hundreds of letters, including to his friend Atticus, and graffiti in Pompeii contained love notes and jokes. The common language of the empire's eastern part was Greek (Koine), and the Septuagint is a Greek translation of the Hebrew Bible. Jesus spoke Aramaic, the language of many people in the Middle East. And the biblical Tower of Babel explains why people speak different languages. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎵 The word "music" comes from the Greek Muses. Literally, it meant...
  - a) the sound of strings
  - b) noise at the market
  - c) birdsong
  - d) the art of the Muses
2. 👁 In hieroglyphs, the names of pharaohs were enclosed in an oval frame called a...
  - a) obelisk
  - b) ankh
  - c) sphinx
  - d) cartouche
3. 🗨 The word "democracy" comes from the Greek words *dēmos* and *kratos*, meaning...
  - a) council of elders
  - b) rule by the people
  - c) law of the city
  - d) voice of the gods
4. ➡ How many letters does the classical Greek alphabet have (from alpha to omega)?
 

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5. ✚ Romans wrote long texts on scrolls that, while reading, had to be...
  - a) unrolled with one hand and rolled up with the other
  - b) spread out flat across the floor
  - c) read aloud while standing in water
  - d) cut into pieces with a knife
6. 🗿 The common language of the eastern part of the Roman Empire was...
  - a) Latin
  - b) Aramaic
  - c) Egyptian
  - d) Greek (Koine)
7. 💖 Cicero left behind hundreds of personal...
  - a) love poems
  - b) drawings
  - c) war diaries
  - d) letters, including to his friend Atticus
8. 🐘 The name "hippopotamus" comes from Greek and means...
  - a) river horse
  - b) big mouth
  - c) water bull
  - d) fat elephant
9. ☆ According to the Bible, the Tower of Babel explains why people...
  - a) worship many gods
  - b) live in cities
  - c) speak different languages
  - d) build houses of brick

1. d 2. d 3. b 4. 24 5. a 6. d 7. d 8. a 9. c

A N S W E R S

# LITERARY STUDIES



## 4. Antiquity

5000 BC – 500 AD

Let's travel back to antiquity, to the oldest great stories. The 'Epic of Gilgamesh' was written on clay tablets in Mesopotamia. Tradition credits the 'Iliad' and the 'Odyssey' to Homer. The 'Iliad' begins with the wrath of Achilles after his quarrel with Agamemnon. The Greeks took Troy by a trick, hiding warriors in a wooden horse. Odysseus wandered home for 10 years; he blinded the Cyclops Polyphemus, who had one eye in the middle of his forehead, and the Sirens lured sailors with their singing. Faithful Penelope wove a shroud and at night unraveled it to delay choosing a suitor. Sappho of Lesbos wrote songs sung to the lyre. In Aesop's fables the heroes are most often animals. Heracles had to perform 12 labors, Theseus escaped the labyrinth thanks to Ariadne's thread, and Medusa turned to stone anyone who looked into her eyes. Sisyphus must forever roll a boulder uphill. In Sophocles, Antigone breaks the king's order to bury her brother, and King Oedipus, learning the truth, blinds himself in despair. Virgil's 'Aeneid' tells of a hero whose descendants founded Rome, and in Ovid's 'Metamorphoses' Daphne turns into a laurel tree. Remember these details – I'll ask you about them in a moment.

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1. 🎵 The poet Sappho of Lesbos wrote songs that were sung to the...
  - a) lyre
  - b) organ
  - c) drum
  - d) trumpet
2. 👁 The Cyclops Polyphemus, blinded by Odysseus, had...
  - a) an eagle's wings
  - b) one eye in the middle of his forehead
  - c) a snake's tail
  - d) three heads
3. 📖 To whom does tradition attribute the Iliad and the Odyssey?
  - a) Virgil
  - b) Hesiod
  - c) Sophocles
  - d) Homer
4. 🗓 How many years did Odysseus spend journeying home after the end of the Trojan War?
 

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5. 🏹 The Greeks captured Troy by a trick, hiding warriors inside...
  - a) wine barrels
  - b) a wooden horse
  - c) hay wagons
  - d) fishing boats
6. 🧑 The Iliad begins with the wrath of Achilles after his quarrel with...
  - a) Hector
  - b) Paris
  - c) Priam
  - d) Agamemnon
7. 💖 While waiting for Odysseus, faithful Penelope wove a shroud by day, and at night she...
  - a) searched for her husband by ship
  - b) prayed in Athena's temple
  - c) unraveled it to delay choosing
  - d) taught her son swordplay
8. 🌿 The main characters in Aesop's fables are usually...
  - a) gods
  - b) animals
  - c) gladiators
  - d) Persian kings
9. ⭐ In Sophocles' tragedy, Antigone defies the king's order in order to...
  - a) become a priestess
  - b) marry a prince
  - c) flee from Thebes
  - d) bury her brother

# CULTURE AND RELIGION

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, to temples, gods and faith. The Greeks had 12 main gods, who lived on Olympus. Apollo's attribute was the lyre, and his priestess at the oracle of Delphi was called the Pythia. The mythical Orpheus with his music moved animals, trees and even stones. The Parthenon was a temple of the goddess Athena, whose sacred bird was the owl. The Dionysia, the festival of Dionysus, gave birth to Greek theater. The Greeks listed 7 wonders of the world. In Rome, the Pantheon was a temple of all the gods, the Vestal Virgins guarded the eternal flame, and the great public baths were the thermae. Jupiter's name is now borne by the largest planet. The Egyptians believed that the heart of the dead would be judged by the gods; they also mummified animals such as cats and ibises, and the god Anubis had the head of a jackal. According to the Bible, Moses led the Israelites out of slavery in Egypt, and the Decalogue is the Ten Commandments. The Buddha reached enlightenment under a fig tree, the bodhi tree. Saint Paul carried Christianity to Athens, Corinth and Rome, among other places, and Saint Augustine described his conversion in his 'Confessions'. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎵 In Greek mythology, which stringed instrument was an attribute of Apollo?
  - a) The aulos
  - b) The tympanon
  - c) The lyre
  - d) The syrinx
2. 👁 The Parthenon on the Athenian Acropolis was a temple of the goddess...
  - a) Hera
  - b) Athena
  - c) Aphrodite
  - d) Demeter
3. 📖 The Decalogue is...
  - a) the ten plagues of Egypt
  - b) the Ten Commandments
  - c) the ten kings of Israel
  - d) the ten books of Homer
4. 🏛 How many wonders of the world did the ancient Greeks list?
 

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5. 🛀 The great public baths in Rome, with pools and exercise halls, were called...
  - a) aqueducts
  - b) thermae
  - c) forums
  - d) basilicas
6. 🏛 The Pantheon in Rome was originally a temple to...
  - a) Emperor Augustus
  - b) Jupiter
  - c) Venus
  - d) all the gods
7. ❤ In which work did St. Augustine describe his path to conversion?
  - a) The City of God
  - b) Summa Theologica
  - c) Confessions
  - d) The Divine Comedy
8. 🌿 The Egyptians mummified not only people but also...
  - a) garden plants
  - b) Nile fish for export
  - c) Roman racehorses
  - d) animals such as cats and ibises
9. ☆ According to the Bible, Moses led the Israelites out of slavery in...
  - a) Persia
  - b) Rome
  - c) Babylon
  - d) Egypt

# ECONOMICS AND FINANCE

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, to merchants, coins and treasures. Before coins, merchants usually exchanged goods for goods. The first coins were struck in the 7th century BC in Lydia from an alloy of gold and silver called electrum, and the rich Lydian king Croesus is still a byword for wealth. The Latin 'pecunia' (money) comes from 'pecus', cattle, and 'salarium' from 'sal', salt. The word 'money' comes from the temple of Juno Moneta, where coins were struck. Athenian tetradrachms showed an owl, the bird of Athena. In Rome, 1 denarius equaled 4 sesterces or 16 asses, so a sesterce was worth 4 asses. The Phoenicians were famous as sailors and merchants; they got purple dye from sea snails (murex), and merchants from Tyre founded Carthage. The Amber Road linked Rome with the Baltic coast. King Midas turned everything to gold, which became a curse because he could not eat or drink; Apollo also gave him donkey's ears because Midas ranked Pan's music above Apollo's. Vespasian, taxing latrines, said 'Pecunia non olet' – money doesn't stink. Seneca wrote that the poor man is the one who craves more. According to the Gospel, Jesus drove the merchants and money changers out of the temple, and Judas betrayed Him for thirty pieces of silver. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎵 According to myth, Apollo gave King Midas donkey ears because Midas...
  - a) ranked Pan's music above Apollo's
  - b) did not pay the musicians
  - c) fell asleep at a concert
  - d) stole his lyre
2. 👁️ Athenian silver coins (tetradrachms) bore the image of...
  - a) a sailing ship
  - b) a lion's head
  - c) an owl, the bird of Athena
  - d) a rose
3. 🗨️ The Latin word "pecunia" (money) comes from "pecus," meaning...
  - a) gold
  - b) cattle
  - c) salt
  - d) land
4. ⚖️ The gift of turning everything into gold became a curse for Midas because...
  - a) the Persians stole it
  - b) gold lost its value
  - c) the gold melted in the sun
  - d) he could not eat or drink
5. 💰 Before coins were introduced, ancient merchants most often...
  - a) paid by card
  - b) used paper banknotes
  - c) exchanged goods for goods
  - d) bought nothing at all
6. 🌐 Phoenician merchants from Tyre founded a colony in North Africa that became a trading power:
  - a) Cairo
  - b) Jerusalem
  - c) Alexandria
  - d) Carthage
7. 💖 Seneca, himself very rich, wrote that the poor man is not the one who has little, but the one who...
  - a) knows no Latin
  - b) has no home
  - c) craves more
  - d) has no children
8. 🌿 The Latin word "salarium" (the source of the English word "salary") comes from "sal," meaning...
  - a) sun
  - b) salt
  - c) grain
  - d) silver
9. ⭐ According to the Gospels, Jesus drove out of the Temple in Jerusalem...
  - a) Roman soldiers
  - b) the sick and the poor
  - c) merchants and money changers
  - d) Pharisees and scribes

# PSYCHOLOGY



## 4. Antiquity

5000 BC – 500 AD

Let's travel back to antiquity to learn how people understood the human soul. The Greek 'psyche' meant soul, breath. The mythical Narcissus fell in love with his own reflection in the water. According to the Bible, young David played the harp to soothe King Saul, tormented by gloom, and Joseph explained Pharaoh's dream of seven fat and seven lean cows. The Pythagoreans used music to calm emotions and heal the soul. Aristotle saw man as a social (political) animal, friendship as one of the greatest goods in life, and placed the seat of thoughts and feelings in the heart; he also noticed that animals can learn by imitation. His student Theophrastus described in his 'Characters' the flatterer and the chatterbox, among others. Hippocrates named four temperaments: sanguine, choleric, melancholic and phlegmatic – optimistic is not one of them. The most famous theater masks are those of tragedy and comedy. The Delphic advice 'nothing in excess' teaches moderation. The Stoic Epictetus taught us to calmly accept what is not up to us, and Seneca advised an angry man to wait before doing anything. Epicurus held that happiness comes from peace of mind and friendship. Roman commanders knew that exercise builds courage and discipline, and the poet Juvenal wrote: 'A sound mind in a sound body'. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏴‍☠️ According to the Bible, young David played the harp in order to...
  - a) scare off the Philistines
  - b) win a contest
  - c) summon rain
  - d) soothe King Saul, tormented by gloom
2. 👁️ In Egypt, Joseph interpreted Pharaoh's dream of seven fat and seven lean...
  - a) camels
  - b) cows
  - c) sheep
  - d) fish
3. 🗨️ The Greek word "psyche," the root of "psychology," originally meant...
  - a) soul, breath
  - b) brain
  - c) dream
  - d) heart
4. ⚖️ Aristotle believed that man is by nature a...
  - a) solitary creature
  - b) social (political) animal
  - c) divine being
  - d) wild beast
5. ✨ The phrase "a sound mind in a sound body" (mens sana in corpore sano) was written by the Roman poet...
  - a) Ovid
  - b) Horace
  - c) Juvenal
  - d) Virgil
6. 🧑‍🤝‍🧑 In the Nicomachean Ethics, Aristotle considered friendship to be...
  - a) unnecessary for the rich
  - b) a danger to the state
  - c) a weakness of wise people
  - d) one of the greatest goods in life
7. 💖 The mythical Narcissus fell in love with...
  - a) a star
  - b) his own reflection in the water
  - c) the goddess Aphrodite
  - d) the nymph Echo
8. 🍃 Aristotle placed the seat of thoughts and feelings in the...
  - a) liver
  - b) heart
  - c) brain
  - d) lungs
9. ⭐ The Stoics, such as Epictetus, taught people to calmly accept whatever...
  - a) the emperor commands
  - b) brings profit
  - c) is pleasant
  - d) is not up to us

# EDUCATION



## 4. Antiquity

5000 BC – 500 AD

Let's travel back to antiquity, to the first schools. In Greece, a 'pedagogue' was a slave who took a boy to school. Athenian education included reading and writing, gymnastics and music, and in the palaestra boys practiced mainly wrestling. The Spartan agoge stressed strength, endurance and discipline. Socrates compared his method to the work of a midwife helping thoughts to be born; he was condemned for allegedly having corrupted the youth. Plato founded his school in the grove of the hero Akademos around 387 BC, and Emperor Justinian closed it in 529 – so it lasted over 9 centuries. Aristotle founded the Lyceum (Lykeion) and described hundreds of animals and their habits. Alexander the Great said he owed his life to his father, and to his teacher the art of living well. Thales calculated a pyramid's height using its shadow. Rhetoric is the art of speaking beautifully and persuasively, and Quintilian advised learning music because it teaches rhythm and voice control. In the Roman ludus, children learned to read, write and count. The trivium included grammar, rhetoric and dialectic (logic). Hypatia of Alexandria was a philosopher and mathematician, and Confucius taught that a good teacher inspires students to think for themselves. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏳️ **The education of boys in Athens included reading and writing, gymnastics and...**
  - a) music
  - b) trade
  - c) sailing
  - d) farming
2. 👁️ **Plato founded his school (c. 387 BC) in a grove dedicated to the hero...**
  - a) Achilles
  - b) Heracles
  - c) Theseus
  - d) Akademos
3. 🗨️ **Rhetoric, taught in Greece and Rome, is the art of...**
  - a) healing
  - b) counting
  - c) building
  - d) speaking beautifully and persuasively
4. 🎵 **The seven liberal arts were divided into the trivium and quadrivium. The trivium included grammar, rhetoric and...**
  - a) music
  - b) arithmetic
  - c) dialectic (logic)
  - d) astronomy
5. 🏹 **The Spartan upbringing of boys (the agoge) placed the greatest emphasis on...**
  - a) strength, endurance and discipline
  - b) sculpture, painting and theater
  - c) poetry, music and dance
  - d) trade, accounting and sailing
6. 🧑 **What was the name of the school Aristotle founded in Athens?**
  - a) The Stoa
  - b) The Lyceum (Lykeion)
  - c) The Academy
  - d) The Garden
7. 💖 **Socrates compared his method of conversation to the work of...**
  - a) a sculptor carving marble
  - b) a soldier capturing a city
  - c) a farmer sowing grain
  - d) a midwife helping thoughts to be born
8. 🌿 **In his History of Animals, Aristotle described...**
  - a) sea monsters
  - b) only horses
  - c) hundreds of animals and their habits
  - d) only insects
9. ⭐ **Hypatia of Alexandria (d. 415) was...**
  - a) a queen of Egypt
  - b) a Roman empress
  - c) a poet from Lesbos
  - d) a philosopher and mathematician

# SOCIOLOGY

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity and look at people in great communities. The first big cities arose in river valleys because there was water and fertile soil there, and Herodotus called Egypt 'the gift of the Nile' because the river's floods deposited fertile silt. The lowest places in Egyptian society belonged to slaves. In Athenian democracy, the vote belonged to adult male citizens, and in ostracism citizens wrote on pottery shards the name of someone to exile. The chorus in Greek theater commented on events with song and dance. In Rome, the 'plebs' meant the common people, and the poor lived in apartment blocks called insulae. The greatest power in the family belonged to the father (pater familias), and Roman women could not vote. At feasts, the rich invited musicians and dancers. The Colosseum held about 50,000 spectators; the retiarius fought there with a net and a trident. Juvenal criticized people who wanted only 'bread and circuses'. A great slave revolt was led by Spartacus, and the philosopher Epictetus was born a slave. The first Christians met in private homes and shared their goods so that no one among them lived in poverty. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏹 The chorus in Greek theater commented on the action on stage through...
  - a) sword fighting
  - b) song and dance
  - c) silence
  - d) war drumming
2. 👁 The Colosseum in Rome could hold about...
  - a) 50,000 spectators
  - b) 5,000 spectators
  - c) 500 spectators
  - d) 500,000 spectators
3. 🗨 The Latin phrase "panem et circenses" means...
  - a) bread and circuses
  - b) peace and war
  - c) gods and men
  - d) law and order
4. ⚖ In Athenian democracy, the right to vote belonged to...
  - a) all residents
  - b) adult male citizens
  - c) priests only
  - d) women and slaves
5. ⚔ A gladiator called a retiarius fought with...
  - a) a chariot
  - b) nothing but his bare hands
  - c) a bow on horseback
  - d) a net and a trident
6. 🧑 Who led the great slave revolt in Italy (73–71 BC)?
  - a) Hannibal
  - b) Spartacus
  - c) Brutus
  - d) Vercingetorix
7. ❤ The Stoic philosopher Epictetus, author of the famous Handbook (Enchiridion), was born a...
  - a) general
  - b) emperor
  - c) slave
  - d) merchant
8. 🌿 The first great cities arose in river valleys because...
  - a) it was easy to defend against floods
  - b) there were no mosquitoes there
  - c) it was colder there
  - d) there was water and fertile soil there
9. ☆ The first Christians shared their property so that...
  - a) no one among them lived in poverty
  - b) they could pay the emperor
  - c) they could buy weapons
  - d) they could build a palace

## LAW



## 4. Antiquity

5000 BC – 500 AD

Let's travel back to antiquity, to the first written laws. The Code of Hammurabi comes from Babylon, and 'an eye for an eye, a tooth for a tooth' is the law of retaliation (lex talionis). According to the Bible, Moses received the tablets of the commandments on Mount Sinai. In Athens, Draco's laws were very harsh, hence 'draconian' penalties. Solon, who was also a poet, abolished slavery for debt. Jurors were chosen by lot, and speeches were timed with a water clock (clepsydra). Socrates refused to escape from prison because he did not want to break his city's laws. In Egypt, according to Diodorus, killing a cat was severely punished, even by death. Roman boys memorized the Law of the Twelve Tables, and important cases were heard in a basilica. Lictors carried the fasces – rods with an ax. Latin maxims teach: 'dura lex, sed lex' – the law is harsh, but it is the law; 'audiatur et altera pars' – let the other side be heard too; 'nulla poena sine lege' – there is no punishment without a law. The sea, the air and running water belonged to all people. In 212 Emperor Caracalla gave citizenship to all free inhabitants of the empire. And Pontius Pilate, who condemned Jesus, was the Roman governor of Judea. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏛️ Cicero recalled that Roman boys learned by heart, like a song, the...
  - a) hymns to Mars
  - b) Law of the Twelve Tables
  - c) odes of Horace
  - d) lists of consuls
2. 👁️ Lictors carried bundles of rods with an ax, a symbol of the power to punish, in front of Roman officials. These were called...
  - a) scepters
  - b) laurels
  - c) fasces
  - d) signa
3. 📖 The Latin maxim "dura lex, sed lex" means...
  - a) no law without punishment
  - b) the law is not retroactive
  - c) the law is harsh, but it is the law
  - d) the law is for everyone
4. ⚖️ The laws of the Athenian lawgiver Draco (7th century BC) were...
  - a) meant only for women
  - b) written in Egyptian on papyrus
  - c) very lenient toward thieves
  - d) very harsh, hence "draconian" penalties
5. ✂️ The principle "an eye for an eye, a tooth for a tooth" is...
  - a) the right to self-defense
  - b) the principle of forgiveness
  - c) the law of retaliation (lex talionis)
  - d) inheritance law
6. 🏰 Which state did the Code of Hammurabi come from?
  - a) Babylon
  - b) Egypt
  - c) Athens
  - d) Persia
7. 💖 The Roman principle "nulla poena sine lege" means that...
  - a) everyone must pay
  - b) the law is for the rich
  - c) the penalty is always death
  - d) there is no punishment without a law
8. 🐈 According to the Greek historian Diodorus, killing a cat in Egypt...
  - a) was rewarded with gold
  - b) was severely punished, even by death
  - c) was allowed only to the pharaoh
  - d) was a duty of the priests
9. ⭐ According to the Bible, Moses received the tablets with the Ten Commandments on Mount...
  - a) Sinai
  - b) Olympus
  - c) Tabor
  - d) Ararat

1. b 2. c 3. c 4. d 5. c 6. a 7. d 8. b 9. a

ANSWERS

# OTHER (HUMANITIES)



## 4. Antiquity

5000 BC – 500 AD

Let's travel back to antiquity, to Muses, wonders and everyday pleasures. Greek mythology named 9 Muses; music and lyric poetry were in the care of Euterpe with her flute. The word 'museum' comes from the Museion of Alexandria, a temple of the Muses and a center of scholars, and the Library of Alexandria collected hundreds of thousands of scrolls from many lands. Alexandria was founded by Alexander the Great. Of the seven wonders of the world, only 1 survives today. At Olympia stood the statue of Zeus, the Colossus of Rhodes depicted the sun god Helios, and the word 'mausoleum' comes from the tomb of Mausolus at Halicarnassus. The Pythian Games were held at Delphi. The god Pan played pipes made of reeds of different lengths. The word 'tragedy' may mean 'song of the goat'. The calendar of 365 days with a leap day every 4 years was introduced in 46 BC by Julius Caesar. Pliny the Elder died in 79 AD while watching the eruption of Vesuvius, and frescoes from Pompeii show gardens with birds, flowers and fruit trees. Roman children played with nuts, dolls and hoops. Queen Zenobia of Palmyra rose up in arms against Rome. The catacombs served Christians mainly as underground cemeteries. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The Muse of music and lyric poetry, shown with a flute, was...
  - a) Euterpe
  - b) Clío
  - c) Melpomene
  - d) Urania
2. 👁 The Colossus of Rhodes, one of the seven wonders of the world, depicted...
  - a) the sun god Helios
  - b) the goddess Athena
  - c) Alexander the Great
  - d) Zeus on his throne
3. 🗨 The word "museum" comes from the Mouseion of Alexandria, which was...
  - a) a pharaoh's royal treasury
  - b) a training school for soldiers
  - c) a temple of the Muses and a center of scholars
  - d) a busy trading port
4. 🏛 How many of the Seven Wonders of the Ancient World survive today?
 

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5. 🏹 The Pythian Games, second only to the Olympics, were held at...
  - a) Troy
  - b) Rome
  - c) Sparta
  - d) Delphi
6. 📖 The Library of Alexandria collected...
  - a) only maps of the night sky
  - b) only books written in Egypt
  - c) hundreds of thousands of scrolls from many lands
  - d) only the laws of the kings
7. ❤ The Roman calendar with a 365-day year and a leap day every 4 years was introduced in 46 BC by...
  - a) Constantine
  - b) Nero
  - c) Julius Caesar
  - d) Augustus
8. 🌿 Pliny the Elder, author of the Natural History, died in AD 79 while observing...
  - a) the eruption of Vesuvius
  - b) a solar eclipse
  - c) a flood of the Tiber
  - d) an earthquake in Greece
9. ⭐ The word "mausoleum" comes from the tomb of King...
  - a) Mausolus at Halicarnassus
  - b) David in Jerusalem
  - c) Ramesses at Thebes
  - d) Alexander in Macedonia

1.a.2.a.3.c.4.15.d.6.c.7.c.8.a.9.a

A N S W E R S

## ARCHITECTURE AND URBAN PLANNING

### 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, to the great builders. Egypt's oldest pyramid, Djoser's at Saqqara, is built as a series of steps. The Great Pyramid of Khufu has a base shaped like a square, and it was the tallest building in the world for about 3,800 years. The ziggurats of Mesopotamia were terraced temple towers, and tradition places the Hanging Gardens in Babylon. The Greek 'acropolis' means high city (a hilltop fortress); on the Athenian Erechtheion stand the caryatids – columns in the shape of women. The style with capitals of acanthus leaves is the Corinthian order, and a slight swelling of columns, entasis, meant that the columns would not look concave. Greek theaters were built on hillsides, and in the theater of Epidaurus a whisper on stage can be heard in the back row. Herostratus set fire to the Temple of Artemis at Ephesus because he wanted to become famous at any cost. Hippodamus of Miletus planned cities on a grid of perpendicular streets. Vitruvius wrote the treatise On Architecture, in ten books. The Roman forum was a square for markets, courts and assemblies, the arch made it possible to build wide spans, and a Pompeian house had an atrium – a hall with an opening in the roof. The Pantheon's dome has at its top an oculus, the 'eye'. Petra in Jordan was carved into red sandstone. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎭 **The theater at Epidaurus is famous because...**
  - a) it has a movable canvas roof
  - b) it was built of cedar wood
  - c) a whisper on stage can be heard in the back row
  - d) it stands partly underwater
2. 👁 **What shape is the base of the Pyramid of Khufu (Cheops) at Giza?**
  - a) A hexagon
  - b) A circle
  - c) A triangle
  - d) A square
3. 📖 **The Roman architect Vitruvius wrote the treatise...**
  - a) The Gallic War
  - b) On Architecture, in ten books
  - c) The Aeneid
  - d) The Republic
4. 🏗 **Hippodamus of Miletus designed cities on a plan of...**
  - a) a grid of perpendicular streets
  - b) a spiral
  - c) concentric circles
  - d) a triangle
5. 🏛 **Builders of Greek temples made columns bulge slightly halfway up (entasis) so that...**
  - a) the roof would be lighter
  - b) more people would fit inside
  - c) the columns would not look concave
  - d) they would be easier to roll
6. 🏛 **The forum in a Roman city was...**
  - a) a track for chariot racing
  - b) a cemetery outside the walls
  - c) the barracks of a legion
  - d) a square for markets, courts and assemblies
7. 💖 **At the top of the Pantheon's dome is a round opening called the...**
  - a) cornice
  - b) capital
  - c) oculus ("eye")
  - d) architrave
8. 🌿 **According to tradition, the Hanging Gardens, one of the wonders of the world, were located in...**
  - a) Carthage
  - b) Athens
  - c) Babylon
  - d) Memphis
9. ☆ **The Temple of Artemis at Ephesus, one of the wonders of the world, was set on fire in 356 BC by Herostratus because he...**
  - a) was fighting the Persian king
  - b) wanted to steal a golden statue
  - c) was a priest of Apollo from Delphi
  - d) wanted to become famous at any cost



#### 4. Antiquity

5000 BC – 500 AD

Let's travel back to antiquity, to the first automatons and the mysteries of electricity. The word 'electricity' comes from the Greek 'elektron', amber – the Greeks noticed that rubbed amber attracts bits of straw and feathers. The stone from Magnesia attracts iron. The Greeks saw lightning as the weapon of Zeus, and the Romans believed a place struck by lightning was sacred and must not be touched. St. Elmo's fire appeared on ships' masts during storms. The doctor Scribonius Largus treated headaches with a live electric ray (a shocking fish). Some think the so-called 'Baghdad battery' could produce a weak current. The word 'automaton' means acting by itself. Ctesibius invented the hydraulis, the ancestor of the organ. Heron of Alexandria built a machine that, for a coin, dispensed a portion of holy water; the aeolipile, spun by steam escaping from nozzles; temple doors that opened when a priest lit a fire on the altar; a mechanical bird that sang using water and air; and a theater of figures moved by a falling weight and strings. The Antikythera mechanism is a gear train for calculating the sky that predicted solar and lunar eclipses. In myth, Hephaestus made a bronze giant, Talos, the guardian of Crete. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎸 The hydraulis, invented by Ctesibius of Alexandria, is the ancestor of the...
  - a) guitar
  - b) drum kit
  - c) trumpet
  - d) organ
2. 👁 The Antikythera mechanism (c. 2nd century BC) is...
  - a) a lock for a temple treasury
  - b) a gear train for calculating the sky
  - c) a sailor's needle compass
  - d) a merchant's bronze scale
3. 🗨 The word "electricity" comes from the Greek "elektron," meaning...
  - a) amber
  - b) lightning
  - c) spark
  - d) metal
4. 🏛 Hero of Alexandria built a machine that, when a coin was dropped in,...
  - a) opened a gate
  - b) dispensed a portion of holy water
  - c) showed the time
  - d) played a melody
5. ⚙ Hero's aeolipile was a sphere that spun thanks to...
  - a) a river current
  - b) the force of the wind
  - c) horses pulling it
  - d) steam escaping from nozzles
6. 🙌 Hero described a device that opened temple doors by itself when a priest...
  - a) clapped his hands
  - b) spoke a password
  - c) lit a fire on the altar
  - d) rang a bell
7. 💖 The Greeks discovered that a stone from Magnesia attracts...
  - a) gold
  - b) wood
  - c) iron
  - d) water
8. 🌿 The ancients observed St. Elmo's fire (to the Romans, the fire of Castor and Pollux) on...
  - a) the tops of pyramids
  - b) fields of grain
  - c) water in wells
  - d) ships' masts during storms
9. ⭐ The Greeks explained lightning bolts as the weapon of...
  - a) Poseidon
  - b) Hermes
  - c) Hades
  - d) Zeus

## COMPUTER SCIENCE AND TELECOMMUNICATIONS

**4. Antiquity**

5000 BC – 500 AD

Let's travel back to antiquity, to the first communication networks and codes. In Africa and Asia, messages were sent by beating drums, and according to Aeschylus the Greeks announced the fall of Troy with a chain of bonfires on mountaintops. Roman soldiers on the frontier warned each other with fire and smoke, and on the battlefield signals were given with horns and trumpets. The lighthouse of Pharos guided ships with fire reflected by mirrors. The Persian relay of messengers on the Royal Road linked Susa with Sardis, and the Roman cursus publicus worked thanks to stations with fresh horses. Alexander the Great sent orders through a system of messengers and signals. Homing pigeons can find their way home from far away. The Caesar cipher works by shifting each letter a few places along the alphabet – with a shift of 3, A becomes D, and C becomes F. The Spartan scytale is a strip wrapped around a rod. Histiaeus hid a message on a slave's shaved head. Polybius wrote letters in a 5 by 5 table, with 25 squares. Romans calculated on a counting board called an abacus, and the Incas recorded data with knotted strings, called quipu. Pliny's 'Natural History' ran to 37 books, and scrolls in libraries were labeled with tags called sillybos. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎶 In ancient Africa and Asia, messages were sent over long distances by...
    - a) plastic mirrors
    - b) beating drums
    - c) radio
    - d) telegraph
  2. 👁 At night, the lighthouse on the island of Pharos near Alexandria guided ships using...
    - a) fire reflected by mirrors
    - b) electric lamps
    - c) flags
    - d) bells
  3. 🗨 The Caesar cipher works by...
    - a) removing the vowels
    - b) shifting each letter a few places along the alphabet
    - c) replacing letters with Roman numerals
    - d) writing from right to left
  4. ➡ In a Caesar cipher with a shift of 3, the letter A becomes D. What does the letter C become? (Latin alphabet)
 

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  5. 📡 Roman soldiers on the frontier passed warnings from tower to tower using...
    - a) letters carried by pigeons
    - b) fire and smoke
    - c) plastic solar mirrors
    - d) shouting
  6. 🗣 According to legend, Alexander the Great sent orders to distant units through...
    - a) messages in bottles
    - b) a system of messengers and signals
    - c) the telephone
    - d) dreams
  7. 💖 The counting board used by the Romans for calculations was the...
    - a) hourglass
    - b) gnomon
    - c) astrolabe
    - d) abacus
  8. 🐦 Carrier pigeons were already used in antiquity because they can...
    - a) fly all night without stopping
    - b) find their way home from far away
    - c) carry heavy parcels
    - d) speak with a human voice
  9. ⭐ The Spartans encrypted messages using a scytale, which was...
    - a) wax seals with a crest
    - b) a strip wrapped around a rod
    - c) a mirror and sunlight
    - d) marks on the messenger's body
- 

1.b2.a3.b4.f5.b6.b7.d8.b9.b

A N S W E R S

# BIOMEDICAL ENGINEERING



## 4. Antiquity

5000 BC – 500 AD

Let's travel back to antiquity, to the first prostheses and medical tools. The word 'surgery' comes from Greek for hand work. Trepanation meant cutting a hole in the skull. According to the Edwin Smith Papyrus, Egyptians set broken bones using splints; they stitched wounds with linen thread and treated deafness by pouring various oil mixtures into the ear. A prosthesis of wood and leather from an Egyptian mummy replaced the big toe, and for sore feet people wore special insoles and dressings. The Etruscans fixed false teeth on gold bridges. Greek doctors listened to the lungs by putting an ear to the chest. The Indian surgeon Sushruta described an operation to reconstruct the nose. In Pompeii over 200 surgical instruments were found, and a speculum from there resembles today's gynecological instruments. Galen studied anatomy on animals such as monkeys and pigs. Romans treated fractures with wooden splints and bandages. The general Marcus Sergius fought on with an iron prosthetic hand. Nero reportedly watched gladiators through an emerald, and Roman women believed crystal balls cool you down in hot weather. The sick left votive offerings in temples of Asclepius – clay models of healed body parts. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏹 Ancient Greek doctors listened to the lungs by pressing against the chest their...
  - a) seashell
  - b) ear
  - c) lyre
  - d) bronze trumpet
2. 👁 More than 200 of these were found in Pompeii:...
  - a) eyeglasses
  - b) thermometers
  - c) surgical instruments
  - d) plastic false teeth
3. 🗨 The word "surgery" comes from Greek and means...
  - a) cutting bones
  - b) study of the body
  - c) hand work
  - d) herbal healing
4. 🏹 Pliny mentions that Nero watched gladiator fights through...
  - a) an emerald
  - b) water in a bowl
  - c) an ice crystal
  - d) a gold plate
5. 🦾 The Roman commander Marcus Sergius, who lost his right hand, kept fighting with...
  - a) a shield tied to his arm
  - b) an iron prosthetic hand
  - c) a wooden leg
  - d) a hook at the elbow
6. 🏹 According to the Edwin Smith Papyrus, Egyptian doctors dressed wounds and...
  - a) transplanted hearts
  - b) set broken bones using splints
  - c) vaccinated children
  - d) took X-rays
7. ❤ As early as the 7th century BC, the Etruscans replaced missing teeth with...
  - a) porcelain teeth
  - b) titanium implants
  - c) teeth made of ice
  - d) false teeth on gold bridges
8. 🌿 Galen studied the structure of the body mainly using...
  - a) marble statues
  - b) animals such as monkeys and pigs
  - c) Egyptian drawings
  - d) the bodies of kings
9. ☆ Trepanation, known since the Stone Age, involved...
  - a) setting a bone
  - b) transfusing blood
  - c) pulling a tooth
  - d) cutting a hole in the skull

# CHEMICAL ENGINEERING



## 4. Antiquity

5000 BC – 500 AD

Let's travel back to antiquity, to workshops of paints, cosmetics and potions. One theory says the word 'chemistry' comes from the Egyptian 'kemet', black earth. Egyptians painted their eyes with kohl made of lead sulfide (galena), and their tombs with paints of minerals ground into powder. The first synthetic pigment, Egyptian blue, was made by heating sand, copper and calcium. Before mummification the body was dried with a salt called natron, and embalmers used resins, oils and natron. Grain was kept in granaries to protect it from moisture and pests. The Latin 'sapo' meant soap – made by boiling fat with wood ash. Smelters producing copper fanned the fire with bellows or pipes. Glassblowing began in Syria and Phoenicia. Roman concrete was durable thanks to volcanic ash (pozzolana). The Greeks crushed olives in presses and made lyre strings from animal gut. Romans ate garum made of fermented fish and salt, sweetened wine with syrup boiled in lead pots, which risked lead poisoning, and Roman women powdered their faces with white lead, which was poisonous. Mithridates VI took small doses of poison to become immune to poisoning. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎶 The ancient Egyptians lined their eyes with black kohl, made among other things from...
  - a) lead sulfide (galena)
  - b) oil-lamp soot and honey
  - c) indigo dye
  - d) elephant bone charcoal
2. 👁 The first synthetic pigment, Egyptian blue, was made by heating...
  - a) eggshells
  - b) sand, copper and calcium
  - c) gold with water
  - d) salt with charcoal
3. 🗞 In Pliny's writings, the Latin word "sapo" (the root of "saponify") meant...
  - a) soap
  - b) wine
  - c) perfume
  - d) glue
4. ⚖ In antiquity, soap was made by boiling fat with...
  - a) flour
  - b) sand
  - c) honey
  - d) wood ash
5. ⚡ To smelt copper from ore, ancient metalworkers...
  - a) soaked it in wine
  - b) ground it into flour
  - c) fanned the fire with bellows or pipes
  - d) froze the ore
6. 🧑‍🔬 Glassblowing on a large scale began in the 1st century BC in...
  - a) Scandinavia
  - b) China
  - c) Syria and Phoenicia
  - d) South America
7. ❤ King Mithridates VI of Pontus reportedly took small doses of poison every day in order to...
  - a) become immune to poisoning
  - b) see better
  - c) lose weight
  - d) become stronger
8. 🌿 Before mummification, the Egyptians dried the body with a natural salt called...
  - a) saltpeter
  - b) lime
  - c) natron
  - d) chalk
9. ⭐ The Romans sweetened wine with a syrup called sapa, boiled in lead pots. Today we know this risked...
  - a) catching a cold
  - b) hearing loss
  - c) color blindness
  - d) lead poisoning

## CIVIL ENGINEERING AND TRANSPORT

**4. Antiquity**

5000 BC – 500 AD

Let's travel back to antiquity, to roads, bridges and sea routes. The Latin 'via' means road, and the saying goes that all roads lead to Rome. The Via Appia ran from Rome to Capua, and later to Brindisi. A Roman road had several layers, topped with flat stones; milestones gave distances in thousands of paces. Legions marched to the rhythm of trumpet signals and footsteps. Aqueducts delivered water without pumps because the channel had a steady, gentle slope, and the Pont du Gard in Gaul has three tiers of arches. Trajan's Bridge over the Danube was for over a thousand years the longest arch bridge in the world. A spoked wheel was lighter and faster than a solid one, and driving a chariot took skill and courage. Egyptians built boats of papyrus and moved pyramid blocks by boat along the Nile and canals. The Phoenicians built ships from Lebanese cedars. The Greek trireme had three rows of oars, and ships were hauled across the Isthmus of Corinth on a stone road called the diolkos. The Pharos lighthouse stood by the harbor of Alexandria. And the Great Wall of China began to be joined into one under Qin Shi Huang. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎖️ **The Via Appia, the most famous Roman road, led from Rome to...**
  - a) Capua, and later to Brindisi
  - b) Athens
  - c) Paris
  - d) London
2. 👁️ **The Pont du Gard aqueduct in Gaul has...**
  - a) three tiers of arches
  - b) wooden spans
  - c) a spiral shape
  - d) a single row of columns
3. 🗨️ **Complete the saying: "All roads lead to..."**
  - a) Babylon
  - b) Troy
  - c) Athens
  - d) Rome
4. ⚙️ **Roman aqueducts delivered water without pumps because...**
  - a) the wind pumped the water
  - b) slaves carried the water
  - c) the channel had a steady, gentle slope
  - d) magnets made water flow uphill
5. 📏 **Milestones along Roman roads showed distances measured in...**
  - a) thousands of paces
  - b) spear lengths
  - c) cubits
  - d) days of marching
6. 🚢 **The Greeks hauled ships across the Isthmus of Corinth on a stone road called the...**
  - a) diolkos
  - b) agora
  - c) via
  - d) stadion
7. ❤️ **The spoked wheel appeared around 2000 BC because, compared with a solid wheel, it was...**
  - a) cheaper in gold
  - b) easier to paint
  - c) quieter at night
  - d) lighter and faster
8. 🌿 **The Egyptians built boats for the Nile out of...**
  - a) bamboo
  - b) iron
  - c) papyrus
  - d) ice
9. 🌟 **The Great Wall of China began to be joined into a single whole during the reign of...**
  - a) Genghis Khan
  - b) Qin Shi Huang
  - c) Alexander the Great
  - d) Julius Caesar

# MATERIALS ENGINEERING

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, to forges and workshops. The Bronze Age is named after an alloy of copper and tin, and it came before the Iron Age because iron requires a higher smelting temperature. Smiths hardened iron by plunging it into water, and the smiths of Damascus and India were famous for patterned, hard steel. The iron pillar of Delhi has barely rusted in 1,600 years. Chinese bianzhong bells were made of bronze, and instruments were made of eight materials, including silk, bamboo and stone. The Silk Road linked Rome with China, and silk was so costly that the Senate tried to forbid men from wearing it. The word 'ceramics' comes from 'keramos', clay; Romans put lamps of terracotta (fired clay) in graves, and Egyptian faience is a blue-green ceramic material. Glass was made by melting sand with soda. Papyrus was made from stems of a plant growing by the Nile, and parchment from treated animal skin. The upper layers of the Pantheon's dome used light pumice. The Greeks painted their statues in bright colors, and bronze sculptures rarely survived because they were often melted down for weapons and coins. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏯 The ancient Chinese bianzhong bells were made of...
  - a) wood
  - b) bronze
  - c) glass
  - d) stone
2. 👁 The Greeks painted their marble statues and temples...
  - a) in bright colors
  - b) only black
  - c) entirely in gold
  - d) only white
3. 🗡 The Bronze Age is named after an alloy of...
  - a) gold and silver
  - b) lead and zinc
  - c) iron and carbon
  - d) copper and tin
4. ⚡ The Bronze Age came before the Iron Age because iron...
  - a) was discovered only in Rome
  - b) requires a higher smelting temperature
  - c) is soft
  - d) does not occur in nature
5. 🔥 Blacksmiths hardened red-hot iron by quickly...
  - a) plunging it into water
  - b) coating it with honey
  - c) burying it in sand
  - d) wrapping it in leather
6. 🍷 In antiquity, glass was made by melting...
  - a) salt with water
  - b) limestone with honey
  - c) sand with soda
  - d) clay with milk
7. 💖 The Romans placed lamps in graves made of...
  - a) acrylic glass
  - b) paper
  - c) terracotta (fired clay)
  - d) rubber
8. 🌿 Papyrus, the Egyptian writing material, was made from...
  - a) calfskin
  - b) birch bark
  - c) limestone
  - d) stems of a plant growing by the Nile
9. ⭐ The Iron Pillar of Delhi (4th–5th century) is famous because...
  - a) it is magnetic
  - b) it was built in Rome
  - c) it glows at night
  - d) it has barely rusted in 1,600 years

# MECHANICAL ENGINEERING



## 4. Antiquity

5000 BC – 500 AD

Let's travel back to antiquity, to the first machines. Archimedes invented a screw for raising water and said: 'Give me a place to stand, and I will move the Earth'. The law of the lever says that 30 kg placed 2 m from the fulcrum balances 20 kg placed 3 m away on the other side. The compound pulley let people lift loads with less force, and the 'claws of Archimedes' grabbed and capsized Roman ships. Archimedes died in Syracuse, telling a soldier: 'Do not disturb my circles!'. Ctesibius improved the water clock, which moved figurines and whistled; in antiquity time was usually measured by sundials and water clocks. The word 'hydraulics' joins 'hydor' and 'aulos', a pipe, and water organs played at games in amphitheaters. Roman cranes were powered by people walking inside a large wheel. Ballistas hurled missiles thanks to twisted ropes made of sinew, and a battering ram is a beam for smashing gates and walls. The watermill used the energy of flowing water, and the sawmill at Hierapolis cut stone with saws driven by a waterwheel. Vitruvius's odometer dropped a pebble after each full turn of the gearing. Daedalus built wings of feathers and wax, and 'deus ex machina' in theater is a sudden resolution of the plot by a god. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏹 **Ctesibius of Alexandria improved the water clock, which...**
  - a) moved figurines and whistled
  - b) powered a boat on the Nile
  - c) showed the wind direction
  - d) baked bread by itself at dawn
2. 🌀 **The block and tackle (compound pulley) allowed the Greeks to...**
  - a) lift loads with less force
  - b) measure time
  - c) count coins
  - d) catch fish
3. 🗨️ **Archimedes said: "Give me a place to stand, and..."**
  - a) I will build a city
  - b) I will move the Earth
  - c) I will stop the Sun
  - d) I will count the stars
4. ⚖️ **A lever: on one side, 30 kg sits 2 m from the pivot. How many kg must be placed 3 m from the pivot on the other side to balance it?**

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5. ⚙️ **Roman construction cranes were powered by...**
  - a) water falling from a canal
  - b) a windmill on a tower
  - c) people walking inside a large wheel
  - d) Hero's steam engine
6. 🏹 **Ballistas and catapults launched projectiles using the energy of...**
  - a) magnets
  - b) gunpowder
  - c) compressed steam
  - d) twisted ropes made of sinew
7. 💖 **The sand hourglass appeared late; in antiquity, time was most often measured with...**
  - a) springs
  - b) wristwatches
  - c) sundials and water clocks
  - d) pendulums
8. 🌿 **The watermill, known since the 1st century BC, ground grain using...**
  - a) the heat of the sun
  - b) the energy of flowing water
  - c) the force of the wind
  - d) the work of elephants
9. ⭐ **A siege battering ram is...**
  - a) a type of boat
  - b) a water clock
  - c) a beam for smashing gates and walls
  - d) a musical instrument



#### 4. Antiquity

5000 BC – 500 AD

Let's travel back to antiquity, to water, mines and energy. According to myth, Prometheus gave people fire. The word 'aqueduct' means conveying water; about a million m<sup>3</sup> a day flowed into Rome through aqueducts, and fountains splashed thanks to water pressure from aqueducts. The Cloaca Maxima was the main sewer. Roman water pipes were made of lead, 'plumbum', which gave English the word 'plumber'. Baths were heated by the hypocaust – hot air under the floor – and they burned so much wood that they led to clearing of nearby forests. Water for Constantinople was stored in underground cisterns. Persian qanats are underground channels that bring water. In Egypt, farmers together built irrigation canals and dams, and measured the Nile with nilometers with a scale. Silver from Laurion let the Athenians build a fleet that beat the Persians. A sentence 'ad metalla' was one of the harshest punishments. Miners heated rocks with fire and then doused them with cold water; at Las Médulas mountains were washed away with water from canals. The Greeks built houses with windows facing south. And salt in Lesser Poland was already being produced in the Stone Age (Neolithic). Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏰 The aqueducts carried into Rome about...

- a) a million m<sup>3</sup> a day
- b) a barrel an hour
- c) 100 buckets a day
- d) 100 liters a week

2. 🗨️ The Cloaca Maxima in Rome is...

- a) a port on the Tiber
- b) the main sewer
- c) the largest circus
- d) the temple of Jupiter

3. 🗨️ The Latin word "plumbum" (lead) is the root of the English "plumber," a worker who fixes...

- a) mine shafts
- b) horseshoes
- c) bread ovens
- d) water pipes

4. ⚖️ The silver mines at Laurion in Attica allowed the Athenians to...

- a) pay the Spartans
- b) build a fleet that beat the Persians
- c) found Rome
- d) build the pyramids

5. 🔥 Ancient miners broke up rocks by heating them with fire and then...

- a) dousing them with cold water
- b) coating them with olive oil
- c) covering them with sand
- d) leaving them for a year

6. 🌿 In Egypt, farmers worked together to build...

- a) irrigation canals and dams
- b) greenhouses
- c) windmills
- d) power plants

7. ❤️ Work in Roman mines was so hard that a sentence "ad metalla" was...

- a) a privilege of citizens
- b) one of the harshest punishments
- c) a reward for gladiators
- d) a profession of priests

8. 🌿 At Las Médulas in Spain, the Romans mined gold by washing away mountains with...

- a) the strength of elephants
- b) water from canals
- c) dynamite
- d) fire and vinegar

9. ☆ According to myth, the Titan Prometheus gave humans...

- a) wind
- b) fire
- c) gold
- d) water

# ASTRONOMY



## 4. Antiquity

5000 BC – 500 AD

Let's travel back to antiquity and look at the stars. The word 'planet' means wanderer; the ancients knew 5 planets visible to the naked eye. The name 'galaxy' comes from the Greek word for milk. The Muse of astronomy, shown with a globe, is Urania. The Pythagoreans believed in the music of the spheres. The zodiac comes from astronomers of Babylon, who recorded observations to predict events in the sky. The Egyptians began their year when Sirius appeared before dawn. Sailors navigated at night by watching the stars. The Greeks named constellations after heroes – Orion is a hunter. Eratosthenes measured Earth's circumference by comparing the length of noon shadows in Syene and Alexandria. Aristarchus of Samos claimed that Earth orbits the Sun, but Ptolemy put the Earth at the center of the universe. Hipparchus created one of the first star catalogs, and the astrolabe was used to measure the altitude of stars. The Chinese recorded comets and sunspots, and the Maya predicted the movements of Venus. Stonehenge is aligned so that at the summer solstice the sun rises over the Heel Stone. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The Pythagoreans believed that the planets make sounds as they orbit, creating...
  - a) the silence of the cosmos
  - b) the music of the spheres
  - c) the Olympic hymn
  - d) the echo of the stars
2. 👁 The zodiac, a belt of constellations along the Sun's path, comes from astronomers in...
  - a) Babylon
  - b) Rome
  - c) China
  - d) the Aztec Empire
3. 🗨 The word "planet" comes from Greek and means...
  - a) fire
  - b) wanderer
  - c) sphere
  - d) goddess
4. 📐 How many planets (visible to the naked eye, other than Earth) did ancient astronomers know?
 

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5. 🌊 At night, ancient sailors navigated by watching...
  - a) the wind from Africa
  - b) birds
  - c) the stars
  - d) the color of the water
6. 🧑 As early as the 3rd century BC, Aristarchus of Samos claimed that...
  - a) the Sun is smaller than the Moon
  - b) stars do not exist
  - c) Earth is flat
  - d) Earth orbits the Sun
7. 💖 Ptolemy placed at the center of the universe...
  - a) the Sun
  - b) the Earth
  - c) the Moon
  - d) Jupiter
8. 🌿 The Egyptians began their year when a certain star appeared before dawn:...
  - a) the North Star
  - b) Aldebaran
  - c) Vega
  - d) Sirius
9. ☆ The Maya predicted the movements of a planet they associated with the god Kukulcan. It was...
  - a) Mars
  - b) Saturn
  - c) Mercury
  - d) Venus

# MATHEMATICS



## 4. Antiquity

5000 BC – 500 AD

Let's travel back to antiquity, to the masters of numbers and shapes. The word 'geometry' means measuring the earth; Egyptian surveyors made right angles with a rope divided into 12 equal sections. The Babylonians solved quadratic equations and counted in a base-60 system, which is why an hour has 60 minutes. Euclid gathered geometry in the Elements, and the angles of a triangle add up to 180 degrees. Pythagoras discovered the laws of harmony by listening to blacksmiths' hammers; his theorem shows that if the legs are 3 and 4, the hypotenuse is 5. The Pythagoreans considered 10 perfect because it is the sum  $1 + 2 + 3 + 4$ . There are five Platonic solids. Archimedes shouted 'Eureka!' when he worked out how to check whether a crown was pure gold, and he placed  $\pi$  between  $3 \frac{10}{71}$  and  $3 \frac{1}{7}$ . The sieve of Eratosthenes shows there are 8 primes less than 20. For larger sums, Romans used pebbles on an abacus; they wrote 49 as XLIX, and MCM is 1900. Our digits come from India, and the Maya were among the first to use a sign for zero. The golden ratio can be seen in shells and plants, and the riddle on Diophantus's tomb lets you work out how long he lived. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 According to legend, Pythagoras discovered the laws of harmony while listening to the sounds of...
  - a) ocean waves
  - b) blacksmiths' hammers
  - c) birdsong
  - d) thunder
2. 👁 How many degrees do the interior angles of a triangle on a plane add up to? (enter a number)
 

-----
3. 🗨 The number 49 written in Roman numerals is...
  - a) IL
  - b) XXXXIX
  - c) XLIX
  - d) LIX
4. 📏 The legs of a right triangle are 3 and 4 long. How long is the hypotenuse?
 

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5. 🗡 Egyptian surveyors ("rope stretchers") marked out right angles using...
  - a) a rope divided into 12 equal sections
  - b) a mirror
  - c) a measuring ruler
  - d) a compass
6. 🌐 The digits we use today (0–9) originated in...
  - a) Greece
  - b) Egypt
  - c) Rome
  - d) India
7. 💖 Archimedes shouted "Eureka!" when, in the bath, he figured out how to...
  - a) check whether a crown was pure gold
  - b) build a catapult
  - c) calculate the distance to the Sun
  - d) measure time
8. 🌿 The Greeks admired the proportion of the "golden ratio," which can be seen, for example, in...
  - a) sand
  - b) clouds
  - c) fire
  - d) shells and plants
9. ☆ Archimedes estimated that the number  $\pi$  lies between...
  - a) 4 and 5
  - b) 3.5 and 4
  - c)  $3 \frac{10}{71}$  and  $3 \frac{1}{7}$
  - d) 2 and 3

1 b 2. 1803. c 4. 55. a 6. d 7. a 8. d 9. c

A N S W E R S

# OTHER (TECHNOLOGY)



## 4. Antiquity

5000 BC – 500 AD

Let's travel back to antiquity, to crafts and everyday inventions. The Greeks believed crafts were given to people by Hephaestus and Athena. From about 3500 BC potters shaped vessels on the potter's wheel, and vases were painted in black-figure and later red-figure style. Greek women wove on looms while singing work songs. Roman mosaics were made of small stones called tesserae. The first fishing nets were made of plant fibers. The first silkworm breeders lived in China; there, around 105 AD, the official Cai Lun invented paper, though the word 'paper' comes from papyrus. The Chinese also invented a compass with a lodestone spoon. Cylinder seals in Mesopotamia were used for 'signing' documents in clay. A water clock measured time by water flowing out of a vessel, and a sundial by the shadow of a rod (gnomon). The Greek aulos is a wind instrument with two pipes. Dice were known in Mesopotamia and the Indus Valley, the Royal Game of Ur is about 4,500 years old, the Egyptian game of senet stood for the soul's journey to the afterlife, and Roman children played games like dice and checkers. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Ancient Greek women wove at their looms while singing...**
  - a) funeral laments
  - b) work songs
  - c) war hymns
  - d) nothing, since silence was required
2. 👁️ **Roman mosaics were made of small pieces called...**
  - a) gems
  - b) tesserae
  - c) bricks
  - d) coins
3. 📜 **The word "paper" comes from...**
  - a) parchment
  - b) bamboo
  - c) papyrus
  - d) pearl
4. ⚙️ **A sundial shows the time by means of...**
  - a) the shadow of a rod (gnomon)
  - b) a burning candle
  - c) moving water
  - d) a weight
5. ⚙️ **From about 3500 BC, potters shaped vessels on a...**
  - a) electric lathe
  - b) potter's wheel
  - c) press
  - d) loom
6. 🧑 **In ancient times, the Chinese invented...**
  - a) eyeglasses
  - b) printing with lead type
  - c) gunpowder for cannons
  - d) a compass with a lodestone spoon
7. 💞 **Cylinder seals in Mesopotamia were used for...**
  - a) "signing" documents in clay
  - b) measuring time in a temple
  - c) grinding grain into flour
  - d) playing dice at the market
8. 🌿 **The first fishing nets were made from...**
  - a) plastic
  - b) plant fibers
  - c) steel wire
  - d) nylon
9. ⭐ **Six-sided dice were already known in...**
  - a) North America
  - b) Rome and nowhere else
  - c) Mesopotamia and the Indus Valley
  - d) only the Middle Ages

1. b 2. b 3. c 4. a 5. b 6. d 7. a 8. b 9. c

ANSWERS

# BIOLOGY

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, when scholars began to look closely at life. The word 'biology' joins the Greek 'bios' and 'logos', life and study, and 'zoology' comes from 'zoon', animal. Aristotle divided animals into those that have blood and those that have none. He noticed that dolphins breathe air and feed their young milk, and he studied embryos by opening hens' eggs on successive days of incubation. According to him and Virgil, bees live in organized communities; beekeepers gathered honey by smoking the hive to calm the bees. Theophrastus, the father of botany, noticed that plants grow toward the light. Pliny the Elder thought the animal most like a human in reason was the elephant. Legend says the poet Arion was saved by a dolphin drawn by his singing. People believed the swan sings beautifully just before it dies – hence 'swan song'. In Aesop's fables, ants stand for hard work and foresight. The Egyptian scarab is a beetle that rolls balls of dung, the ibis was the bird of Thoth, god of wisdom and writing, and the phoenix Bennu a symbol of rebirth from the ashes. Egyptians caught birds in the marshes with throwing sticks. Many peoples also believed in spontaneous generation, for example mice from Nile mud. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏹 According to Greek legend, the poet Arion, thrown overboard from a ship, was saved by...
  - a) a whale
  - b) a shark
  - c) a dolphin drawn by his singing
  - d) a sea turtle
2. 👁 The sacred scarab of the Egyptians is actually...
  - a) a desert spider
  - b) a grasshopper
  - c) a beetle that rolls balls of dung
  - d) a moth
3. 🗨 The word "biology" combines the Greek "bios" and "logos," meaning...
  - a) body and motion
  - b) earth and water
  - c) animal and plant
  - d) life and study
4. 🐬 Aristotle noticed that dolphins...
  - a) live only in freshwater rivers
  - b) have gills like other fish
  - c) breathe air and feed their young milk
  - d) lay eggs in warm sand
5. 🐝 Ancient beekeepers collected honey after first...
  - a) singing loudly
  - b) heating it with fire
  - c) pouring water on the hive
  - d) smoking the hive to calm the bees
6. 🐝 According to Aristotle and Virgil, bees live in...
  - a) organized communities
  - b) solitude
  - c) pairs
  - d) herds with bumblebees
7. ❤ Theophrastus, a student of Aristotle, is called the father of...
  - a) botany
  - b) astronomy
  - c) zoology
  - d) medicine
8. 🌿 Pliny the Elder believed that the animal most like humans in intelligence was the...
  - a) lion
  - b) elephant
  - c) horse
  - d) eagle
9. ☆ The Egyptian phoenix (Bennu) was a symbol of...
  - a) rebirth from the ashes
  - b) famine
  - c) war
  - d) the sea

1. c 2. c 3. d 4. c 5. d 6. a 7. a 8. b 9. a

A N S W E R S

# CHEMISTRY

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, to the workshops of smiths, dyers and alchemists. Empedocles believed everything is made of four elements: earth, water, air and fire, and Democritus called the smallest particles of matter atoms. The ancients knew seven metals: gold, silver, copper, iron, tin, lead and mercury; the Greeks called mercury silver water. Gold was seen as the metal of the gods because it never rusts; the Latin 'aurum' gave the symbol Au. Goldsmiths purified it by melting it with lead in crucibles, and the alchemists of Alexandria dreamed of turning ordinary metals into gold. Bronze was used for the sistrum, the priestesses' rattle, and for bells – the more tin, the harder and more resonant the alloy. Imperial purple was dyed with dye from sea snails. Egyptians brewed beer by fermenting barley bread with water. Cleopatra, the story goes, dissolved a pearl in vinegar. Romans got salt by evaporating seawater in shallow pools, and kept snails and oysters in salt water. Frankincense and myrrh came from Arabia and East Africa, and sulfur came from near volcanoes. In 79 AD Vesuvius buried Pompeii under a layer of ash and pumice. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏹 **Ancient bells were cast from bronze. The more tin it contains, the more the alloy is...**
  - a) hard and resonant
  - b) soft like lead
  - c) transparent
  - d) lighter than water
2. 👁 **The Greeks called mercury, the only metal that is liquid at room temperature,...**
  - a) silver water
  - b) golden fire
  - c) black stone
  - d) white sand
3. 🗨 **The Latin name for gold, "aurum," gave us the chemical symbol...**
  - a) Go
  - b) Gd
  - c) Ar
  - d) Au
4. ⚖ **The ancients knew seven metals: gold, silver, copper, iron, tin, lead and...**
  - a) uranium
  - b) mercury
  - c) titanium
  - d) aluminum
5. 🔥 **To purify gold, ancient goldsmiths...**
  - a) froze it
  - b) boiled it in milk
  - c) soaked it in wine
  - d) melted it with lead in crucibles
6. 🐚 **According to an anecdote, Cleopatra won a bet with Mark Antony by dissolving in vinegar a...**
  - a) pearl
  - b) gold coin
  - c) diamond
  - d) emerald
7. 💖 **The alchemists of Alexandria (such as Zosimos) dreamed of turning ordinary metals into...**
  - a) water
  - b) glass
  - c) stone
  - d) gold
8. 🌿 **Sulfur, used in antiquity to fumigate houses, came most often from areas near...**
  - a) ice deserts
  - b) forests
  - c) rivers
  - d) volcanoes
9. ⭐ **The ancients considered gold the metal of the gods because it...**
  - a) glows at night
  - b) is liquid
  - c) is the heaviest
  - d) never rusts

# PHYSICS

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, to the birth of physics – the word comes from the Greek 'physis', nature, and 'energy' from 'energeia', activity. Heraclitus said you cannot step into the same river twice. Pythagoras studied sounds on a monochord, an instrument with a single string, and Aristotle explained that sound travels through vibrations of the air. Aristotle also taught that heavier bodies fall faster – this was disproved only by Galileo. Archimedes stated his law: a body in a liquid seems to lose as much weight as the liquid it displaces; he also proved that a lever balances when the products of weight and arm on both sides are equal. Legend says he set Roman ships on fire with mirrors focusing the sun's rays. The Roman steelyard, the statera, had unequal arms and a sliding weight. Euclid thought rays of sight come out of the eyes, and Ptolemy studied how light changes direction passing from air into water – that is why a spoon in a glass looks broken. The Greeks knew the Moon shines with light reflected from the Sun, and that Earth's shadow during an eclipse is round, which proves Earth is a sphere. Anaxagoras was accused of impiety for saying the Sun is a red-hot stone. Lucretius taught that the world arose from the motion and joining of atoms. And Roman soldiers in the 'tortoise' formation protected themselves with shields held over their heads. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Pythagoras studied sounds on a monochord, an instrument with...**
  - a) three pipes
  - b) two drums
  - c) a hundred strings
  - d) a single string
2. 🏹 **According to legend, Archimedes set Roman ships on fire at Syracuse using...**
  - a) catapults with torches
  - b) mirrors focusing the sun's rays
  - c) arrows with gunpowder
  - d) Greek fire
3. 🗨️ **The word "physics" comes from the Greek "physis," meaning...**
  - a) nature
  - b) force
  - c) body
  - d) motion
4. ⚖️ **Archimedes' principle: a body immersed in a liquid seems to lose as much weight as the weight of...**
  - a) the liquid it displaces
  - b) the body itself
  - c) the air above the liquid
  - d) the vessel of liquid
5. 🛡️ **In the testudo ("tortoise") formation, Roman soldiers protected themselves from arrows...**
  - a) behind a wall
  - b) with shields held over their heads
  - c) in trenches
  - d) under wagons
6. 🧑 **Aristotle taught that heavier bodies fall faster. This was finally disproved by...**
  - a) Copernicus
  - b) Einstein
  - c) Newton
  - d) Galileo
7. 💧 **A spoon placed in a glass of water appears...**
  - a) broken
  - b) invisible
  - c) longer
  - d) turned to gold
8. 🌑 **The Greeks noticed that during a lunar eclipse, Earth's shadow is...**
  - a) invisible
  - b) square
  - c) round, which proves Earth is a sphere
  - d) blood red
9. ⭐ **In his poem On the Nature of Things, Lucretius taught that the world came into being through...**
  - a) a cosmic egg
  - b) the will of the Olympian gods
  - c) the motion and joining of atoms
  - d) the tears of the Titans

# ENVIRONMENT

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity and look at land, seas and volcanoes. Mount Olympus, home of the gods, is the highest peak in Greece. The Greeks believed the sound of the wind was the voice of the god Aeolus, and explained Etna's eruptions as the work of Hephaestus's forge. The Romans called the Mediterranean 'Mare Nostrum', our sea. Athens built the Long Walls to Piraeus to bring in food by sea during a siege. Greek fishermen gathered sponges by diving while holding their breath, with a stone. Plato described how, after forests in Attica were cut down, the soil washed away, leaving bare hills; he also described Atlantis, which sank into the sea in a single day and night. The eruption on Santorini may have contributed to the fall of the Minoan civilization on Crete. Pompeii lies at the foot of Vesuvius, and in 62 AD its people lived through a strong earthquake. The name 'Mesopotamia' means land between rivers; the fall of some of its cities is linked to soil salinization caused by irrigation. About 6,000 years ago the Sahara, the largest desert on the caravan routes, began drying out, so people moved to the Nile. China's Dujiangyan irrigation system still works today. Horace wrote 'carpe diem' – seize the day, and Virgil sang of shepherds in his 'Eclogues'. Remember these details – I'll ask you about them in a moment.

---

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Mount Olympus, the home of the gods, is the highest peak in...**
  - a) Egypt
  - b) Greece
  - c) Turkey
  - d) Italy
2. 🗺️ **The largest desert crossed by caravan routes heading west from Egypt is the...**
  - a) Kalahari
  - b) Atacama
  - c) Gobi
  - d) Sahara
3. 🗞️ **The Romans called the Mediterranean "Mare Nostrum," meaning...**
  - a) our sea
  - b) great sea
  - c) sea of the gods
  - d) warm sea
4. 🏰 **About 6,000 years ago the Sahara began to dry out, so people...**
  - a) built ports there
  - b) raised hippos there
  - c) moved to the Nile
  - d) planted forests
5. 🧜♂️ **Greek fishermen gathered sponges by diving...**
  - a) with goatskin air tanks
  - b) in wooden submarines
  - c) holding their breath, with a stone
  - d) in steel helmets with a tube
6. 🏞️ **In the Critias, Plato described how, after the forests of Attica were cut down,...**
  - a) volcanoes appeared
  - b) the sea dried up
  - c) soil washed away, leaving bare hills
  - d) the climate cooled
7. 💖 **The Roman poet Horace valued simple country life and wrote the famous words "carpe diem," meaning...**
  - a) listen to the wind
  - b) seize the day
  - c) tend your garden
  - d) love nature
8. 🌿 **The legendary Atlantis, described by Plato, supposedly sank into...**
  - a) the desert sands
  - b) a volcano's lava
  - c) the ice
  - d) the sea in a single day and night
9. ⭐ **The eruption on Santorini (Thera) around 1600 BC may have contributed to the decline of the...**
  - a) Persian civilization
  - b) Egyptian civilization
  - c) Minoan civilization on Crete
  - d) Roman civilization

# HEALTH

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, to the roots of caring for health. The word 'hygiene' comes from Hygieia, goddess of health and daughter of Asclepius. The rod of Asclepius is a staff with a snake coiled around it. In the temples of Asclepius, the sick were treated with music and sleep in the temple, and lyres were played and paeans sung because music was believed to aid healing. Hippocrates held that diseases have natural causes, not divine ones. He is credited with 'primum non nocere' – first, do no harm; his oath binds doctors to keeping patients' secrets. He recommended rest, a good diet and fresh air, and advised building towns away from marshes, where fevers are common. Celsus advised healthy people to live actively, without rigid rules, and Roman doctors believed that sleep, moderation and walks protect your health. The Greeks exercised in gymnasiums naked. In the baths, people passed through a hot room, a warm room and a cold room, and latrines were shared, with many seats. Teeth were cleaned with powder of ash and ground bones or shells. Egyptians protected themselves from mosquitoes by sleeping under nets. The plague of Athens killed Pericles, and the Antonine Plague was most likely smallpox. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏹 In the temples of Asclepius, the Greeks treated insomnia, among other ways, with...
  - a) bloodletting
  - b) work in a mine
  - c) a week of fasting
  - d) music and sleep in the temple
2. 👁 The Rod of Asclepius, the symbol of medicine, is a staff entwined by a...
  - a) snake
  - b) grapevine
  - c) chain
  - d) pair of doves
3. 🗨 The principle "primum non nocere" is attributed to Hippocrates. It means...
  - a) treat first
  - b) the doctor always knows
  - c) health is a blessing
  - d) first, do no harm
4. ⚖ Hippocrates argued that diseases have...
  - a) causes in dreams
  - b) natural causes, not divine ones
  - c) purely magical causes
  - d) causes only in the stars
5. ⚔ Greeks exercised in the gymnasium...
  - a) in armor
  - b) naked
  - c) in long robes
  - d) in hobnailed boots
6. 🧑 Among other things, the Hippocratic Oath commits a doctor to...
  - a) treating the rich for free
  - b) fighting in the army
  - c) keeping patients' secrets
  - d) praying to Zeus every day
7. ❤ The Roman physician Celsus advised that a healthy person should...
  - a) live actively, without rigid rules
  - b) avoid the sun completely
  - c) eat only fresh fruit
  - d) sleep as much as possible by day
8. 🍃 The Antonine Plague of the 2nd century devastated the Roman Empire. Most researchers believe it was...
  - a) bubonic plague
  - b) smallpox
  - c) bird flu
  - d) cholera
9. ☆ Roman doctors believed that good sleep, moderate eating and walks...
  - a) protect your health
  - b) are for slaves
  - c) weaken soldiers
  - d) shorten your life

1. d 2. a 3. d 4. b 5. b 6. c 7. a 8. b 9. a

ANSWERS

# PHYSICAL CULTURE AND SPORT

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, to stadiums and arenas. The Olympic Games were held in Olympia from 776 BC every 4 years, so the next ones fell in 772 BC. For the Games a sacred truce between the cities was declared. Athletes swore they had trained honestly for 10 months, practiced on soft sand, and before a bout rubbed their bodies with olive oil. Winners received a wreath of olive branches and at home were welcomed as a hero. A stadion measured about 190 meters, and the long race, the dolichos, about 24 stadia – roughly 5 kilometers. The five-event contest was called the pentathlon, long jumpers kept rhythm to the sound of the aulos, and in the pankration only biting and eye-gouging were forbidden. Milo of Croton built his strength by carrying a growing calf every day. Myron's sculpture 'Discobolus' shows an athlete throwing a discus. At the Pythian Games in Delphi, people also competed in lyre playing and singing. In Rome, chariot races were held in the Circus Maximus, and fans cheered teams marked by team colors. The Olympic Games ended around 393 after a ban by Emperor Theodosius I. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏳️ At the ancient games, long jumpers kept the rhythm of their jump to the sound of the...
  - a) trumpets
  - b) lyre
  - c) aulos
  - d) drums
2. 👁️ Winners of the Olympic Games received a wreath made of...
  - a) gold
  - b) roses
  - c) Delphic laurel
  - d) olive branches
3. 🏠 The Olympic five-event contest was called in Greek the...
  - a) stadion
  - b) pentathlon
  - c) pankration
  - d) marathon
4. 🏊 The ancient Olympic Games were held every 4 years starting in 776 BC. In what year BC were the next Games after 776?
 

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5. 🏹 Where were the ancient Olympic Games held?
  - a) In Sparta
  - b) In Olympia
  - c) In Rome
  - d) In Athens
6. 🧑 For the duration of the Games at Olympia, the Greeks proclaimed...
  - a) a sacred truce between the cities
  - b) a new king
  - c) a war tax
  - d) a ban on trade
7. ❤️ According to legend, Milo of Croton, the famous wrestler, built up his strength by every day...
  - a) carrying a growing calf
  - b) lifting stones from a river
  - c) running at night
  - d) swimming in the sea
8. 🍃 Before competing, ancient athletes rubbed their bodies with...
  - a) honey
  - b) Nile mud
  - c) milk
  - d) olive oil
9. ⭐ The Olympic Games stopped being held around AD 393 after a ban by Emperor...
  - a) Nero
  - b) Theodosius I
  - c) Augustus
  - d) Julian

1 c 2 d 3 b 4 772 5 b 6 a 7 a 8 d 9 b

A N S W E R S

# MEDICINE

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, to the first great physicians. The Egyptian Ebers Papyrus contains hundreds of recipes for medicines. Egyptians put honey, which fights bacteria, on wounds, as well as moldy bread – today we know mold can inhibit the growth of bacteria. The architect and physician Imhotep was worshipped after death as a god of medicine. The word 'diagnosis' means recognition, and 'anatomy' means cutting up. Hippocrates taught that health depends on the balance of the four humors (fluids) in the body, advised watching the patient's face, breathing and sleep, and wrote: 'Life is short, art is long'. Herophilus of Alexandria found that the center of thought is the brain, and compared the pulse to rhythm in music. Galen was a doctor to gladiators and later to Emperor Marcus Aurelius; he discovered that arteries carry blood, not air. Celsus described cataract surgery by pushing the lens aside with a needle. Roman army doctors stitched wounds and stopped bleeding with tourniquets, and Asclepiades of Bithynia treated patients with music and baths. The evangelist Luke was a physician by profession, and the brothers Cosmas and Damian healed people free of charge. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The Greek physician Herophilus studied the pulse, comparing its rhythm to...
  - a) a soldier's march
  - b) the flight of birds
  - c) rhythm in music
  - d) the motion of waves
2. 👁 The oldest Egyptian medical papyri, such as the Ebers Papyrus, contain...
  - a) war hymns
  - b) star maps
  - c) tax tables
  - d) hundreds of recipes for medicines
3. 🗨 The word "diagnosis" comes from Greek and means...
  - a) medicine
  - b) recognition
  - c) pain
  - d) surgery
4. ⚖ Hippocrates taught that health depends on the balance of four...
  - a) planets
  - b) humors (fluids) in the body
  - c) elements in the home
  - d) times of day
5. ✚ Roman army doctors stitched wounds and...
  - a) used ether as general anesthesia
  - b) transplanted limbs
  - c) stopped bleeding with tourniquets
  - d) removed hearts
6. 🧑 Galen was a doctor to gladiators in Pergamon, and later to...
  - a) the pharaoh
  - b) Emperor Marcus Aurelius
  - c) Alexander the Great
  - d) the king of Persia
7. ❤ Hippocrates wrote: "Life is short, art is..."
  - a) eternal
  - b) difficult
  - c) long
  - d) beautiful
8. 🌿 The Egyptians applied to wounds...
  - a) honey, which fights bacteria
  - b) desert sand
  - c) fresh meat
  - d) papyrus leaves
9. ☆ Luke, the author of one of the Gospels, was by profession a...
  - a) fisherman
  - b) physician
  - c) tax collector
  - d) carpenter

# PHARMACY

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, to the first pharmacies and herbs. The Greek 'pharmakon' meant medicine, but also poison, because the Greeks knew that a drug and a poison differ mainly in dose. Dioscorides described about 600 medicinal plants, and Roman herbals showed them in colored drawings. Apothecaries ground herbs in a mortar. Galen prepared theriac, an antidote to poisons, for the emperors. Poppy juice was used to relieve pain and induce sleep, and willow bark contains a substance similar to aspirin. Sage's Latin name, 'salvia', comes from 'salvare', to save; mint, chamomile and sage were used by Romans as medicinal herbs. Sore joints got poultices of hot clay and herbs. From India came pepper, cinnamon and other spices. Garlic was given to pyramid builders because it was believed to give strength and protect against illness. In myth, Circe turned Odysseus's companions into pigs with a magic potion, and Helen's nepenthe brought forgetfulness of sorrow. Cleopatra reportedly died from the bite of a cobra (asp). Socrates was executed by drinking hemlock; before dying he asked for a rooster for Asclepius, because he saw death as a 'cure' for life. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏹 The mythological sorceress Circe turned Odysseus's companions into pigs using...  
a) singing  
b) a fishing net  
c) a black stone  
d) a magic potion
2. 👁 Roman herbals depicted medicinal plants in...  
a) DVDs  
b) statues  
c) colored drawings  
d) photographs
3. 🗨 The word "pharmacy" comes from the Greek "pharmakon," meaning...  
a) drink of the gods  
b) medicine, but also poison  
c) doctor's hand  
d) herb garden
4. 🍷 Socrates was executed by being made to drink...  
a) water from the Lethe  
b) wine with honey  
c) hemlock  
d) poppy juice
5. 🔥 Roman apothecaries ground herbs in a...  
a) watermill  
b) furnace  
c) kettle over a fire  
d) mortar
6. 🧑 Theriac, the famous antidote to poisons, was prepared for emperors by...  
a) Hippocrates  
b) Archimedes  
c) Pythagoras  
d) Galen
7. 💖 The Greeks used poppy juice to...  
a) relieve pain and induce sleep  
b) write  
c) dye fabrics  
d) cook soup
8. 🍃 Willow bark, used by Hippocrates for pain and fever, contains a substance similar to...  
a) aspirin  
b) insulin  
c) penicillin  
d) vitamin C
9. ⭐ According to ancient authors, Cleopatra died from the bite of a...  
a) spider  
b) shark  
c) cobra (asp)  
d) scorpion

# AGRICULTURE

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, to the fields and granaries. The Fertile Crescent covered Mesopotamia, Syria and Palestine; the oldest grains grown there were wheat and barley. Rice was first grown in China, and maize was domesticated in Mexico. Egyptians divided the year into three seasons: flood, sowing and harvest; they lifted water with a shaduf, a pole lift with a counterweight, and at harvest they sang songs to the rhythm of work. Grain was threshed by having animals trample the ears. The Greeks grew grapes and olives because they tolerate dry, hot summers well; their goddess of harvest was Demeter. Roman plows were usually pulled by oxen, and in crop rotation a field was left fallow every other year. Sicily and Egypt, 'Rome's granary', supplied grain, and the 'Bread of Rome' came mainly from Egypt. Cato the Elder advised in 'On Agriculture' on running a farm with slaves, and wrote that a good farmer sells more than he buys. Virgil described farm work in the Georgics. The Saturnalia were celebrated in December, after sowing. Remember these details – I'll ask you about them in a moment.

---

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏳️ **The Roman poet Virgil described farm work in his poem...**
  - a) the Aeneid
  - b) the Metamorphoses
  - c) the Georgics
  - d) the Odes
2. 👁️ **The Fertile Crescent, the cradle of agriculture, included...**
  - a) the Sahara
  - b) Central Europe
  - c) Siberia
  - d) Mesopotamia, Syria and Palestine
3. 🗨️ **The Romans called Sicily and Egypt "the granary of Rome" because they supplied...**
  - a) wine
  - b) grain
  - c) marble
  - d) slaves
4. 🗓️ **The Egyptians divided the year into three seasons tied to the Nile:**
  - a) rain, wind and heat
  - b) winter, spring and summer
  - c) flood, sowing and harvest
  - d) war, peace and festivals
5. 🌾 **The ancients threshed grain by having...**
  - a) animals trample the ears
  - b) people eat them
  - c) the wind scatter them
  - d) women cook them
6. 🗿 **In his work On Agriculture, Cato the Elder gave advice on...**
  - a) running a farm with slaves
  - b) fighting Carthage
  - c) building ships
  - d) writing poetry
7. ❤️ **The Greeks grew grapevines and olives because these plants...**
  - a) tolerate dry, hot summers well
  - b) bloom in winter
  - c) grow only in the mountains
  - d) need a lot of rain
8. 🌱 **Rice was first cultivated in...**
  - a) Greece
  - b) China
  - c) Gaul
  - d) Egypt
9. ⭐ **The Greek goddess of agriculture and harvest was...**
  - a) Artemis
  - b) Demeter
  - c) Athena
  - d) Hera

# VETERINARY SCIENCE

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, to the first animal doctors. The Latin 'veterinarius' referred to the care of pack and draft animals. Roman 'mulomedici' treated mainly mules and horses, and for a lame horse they examined its hooves and applied dressings. Hooves were protected with iron hipposandals. The Hippiatrica is a Byzantine collection of texts on treating horses. Xenophon advised training a horse gently, not in anger, and army horses were taught to respond to trumpet signals and got used to the noise of battle and trumpets. Bucephalus was the horse of Alexander the Great; he was afraid of his own shadow, so Alexander turned him to face the sun. In honor of his horse, Alexander founded the city of Bucephala, and named another after his dog Peritas. Caligula reportedly wanted to make his horse Incitatus a consul. In the 'Odyssey', the dog Argos alone recognized Odysseus after 20 years. The oldest images of pet dogs come from Egyptian tombs, and when a cat died, Egyptians mourned by shaving their eyebrows. Aristotle described rabies in dogs. The lion-headed goddess Sekhmet was a goddess of war and disease, but also healing. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏹 **War horses in Greece and Rome were trained to respond to...**
  - a) birds' whistling
  - b) trumpet signals
  - c) the smell of herbs
  - d) the color of a banner
2. 👁 **Bucephalus was the famous horse of...**
  - a) Alexander the Great
  - b) Hannibal
  - c) Caligula
  - d) Julius Caesar
3. 🗨 **The Latin word "veterinarius" referred to caring for...**
  - a) ships
  - b) gardens
  - c) children
  - d) pack and draft animals
4. ➡ **Bucephalus was afraid of his own shadow, so young Alexander...**
  - a) turned him to face the sun
  - b) rode him at night
  - c) covered his eyes
  - d) sold him
5. ⚔ **The Romans protected horses' hooves with iron sandals called...**
  - a) nailed horseshoes
  - b) spurs
  - c) stirrups
  - d) hipposandals
6. 🗣 **According to an anecdote, Caligula loved his horse Incitatus so much that he wanted to make him a...**
  - a) king of Egypt
  - b) priest
  - c) gladiator
  - d) consul
7. ❤ **In his work On Horsemanship, the Greek author Xenophon advised training a horse...**
  - a) by beating it for every mistake
  - b) in darkness
  - c) gently, not in anger
  - d) without feeding it before a race
8. 🍀 **In the Odyssey, the dog Argos was the only one who...**
  - a) guarded the temple
  - b) swam across the sea
  - c) defeated the Cyclops
  - d) recognized Odysseus after 20 years
9. ☆ **In Egypt, when a household cat died, the family showed their mourning by...**
  - a) shaving their eyebrows
  - b) giving up meat forever
  - c) painting the walls
  - d) burning the house



## ANIMAL SCIENCE AND FISHERIES

### 4. Antiquity

5000 BC – 500 AD

Let's travel back to antiquity, to herders and fishers. Goats and sheep were domesticated in the Middle East about 10,000 years ago. The domestic chicken descends from the red junglefowl of Asia. Egyptians domesticated the donkey for carrying loads, and camels made caravans across the desert possible. Arabian horses are famous for endurance and speed. The sacred bull Apis was worshipped as an incarnation of the god Ptah. Fishermen on the Nile fished with nets, harpoons and baskets. Romans raised fish in ponds called piscinae, Sergius Orata became famous for farming oysters, and in gliraria they fattened dormice (small rodents). Singing nightingales were sold to the rich, and peacocks were brought from India. Juno's sacred geese saved Rome when they warned of the Gauls by honking. Columella advised breeders to care for animals' health as for their own. 'Apicultura' comes from 'apis', bee, and silkworms are fed mulberry leaves. For the Greeks, the cicada was the symbol of summer and poetry, and the Greek 'ichthys', fish, became a secret sign of the first Christians. Remember these details – I'll ask you about them in a moment.

---

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🦗 Which singing insect was a symbol of summer and poetry for the Greeks?
  - a) The ant
  - b) The cicada
  - c) The bee
  - d) The moth
2. 👁 The Romans raised fish in special ponds called...
  - a) horrea
  - b) castra
  - c) thermae
  - d) piscinae
3. 🗨 The Greek word "ichthys" (fish) was a secret sign of the first...
  - a) Christians
  - b) gladiators
  - c) merchants
  - d) sailors
4. 🐪 Camels were domesticated around 1000 BC on the Arabian Peninsula, which made possible...
  - a) wars in the snow
  - b) caravans across the desert
  - c) chariot races
  - d) seafaring
5. 🦎 The Roman entrepreneur Sergius Orata became famous for farming...
  - a) parrots
  - b) elephants
  - c) snakes
  - d) oysters
6. 🐐 Goats and sheep were domesticated in the Middle East about...
  - a) 10,000 years ago
  - b) 1,000 years ago
  - c) in Roman times
  - d) 100,000 years ago
7. ❤ Arabian horses are famous for their...
  - a) great height
  - b) endurance and speed
  - c) swimming
  - d) heavy build
8. 🐔 The domestic chicken is descended from the wild...
  - a) ostrich of Africa
  - b) red junglefowl of Asia
  - c) turkey of America
  - d) pheasant of Poland
9. ⭐ According to legend, Juno's sacred geese saved Rome when they...
  - a) found water
  - b) ate the locusts
  - c) showed the way
  - d) warned of the Gauls by honking

# OTHER (NATURE)

## 4. Antiquity

5000 BC – 500 AD



Let's travel back to antiquity, to travelers and explorers of nature. The word 'geography' was coined by Eratosthenes – it means describing the Earth; his map had lines resembling meridians and parallels. Strabo described in 17 books the lands of the known world. Pytheas of Massalia linked the tides to the Moon. According to Herodotus, Phoenicians sent by Pharaoh Necho sailed around Africa, and Hanno of Carthage described 'wild people' who were probably gorillas or chimpanzees. Sailors waited for the season of mild summer winds. The Greeks imagined Oceanus as a great river encircling the world. Pliny thought amber was fossilized tree resin. The violet amethyst was said to protect against drunkenness. Romans cooled drinks with snow brought down from the mountains. Thales supposedly fell into a well while gazing at the stars. Seneca wrote that admiring nature calms the mind and teaches humility. One of the Colossi of Memnon 'sang' at dawn, and 'singing' sands make sound when grains slide down a dune. In myth, the nymph Echo could only repeat other people's words, the myth of Phaethon explained how deserts arose, and at the bottom of Pandora's box hope remained. Remember these details – I'll ask you about them in a moment.

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Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 Today we know that "singing" sands make a sound when...
  - a) grains slide down a dune
  - b) a river flows underground
  - c) the wind plays on shells
  - d) the sun warms them
2. 👁️ The ancient Greeks knew a purple stone that was believed to protect against drunkenness:
  - a) ruby
  - b) sapphire
  - c) diamond
  - d) amethyst
3. 🗨️ The word "geography" was coined by Eratosthenes. It means...
  - a) measuring the sky
  - b) making sea charts
  - c) the study of rocks
  - d) describing the Earth
4. 🌊 Pytheas of Massalia (c. 325 BC) was one of the first to link ocean tides to...
  - a) the wind
  - b) the Sun alone
  - c) volcanoes
  - d) the Moon
5. ⚡ To cross the Mediterranean, sailors waited for...
  - a) winter storms
  - b) the full moon
  - c) the season of mild summer winds
  - d) a solar eclipse
6. 🧑 Herodotus described a voyage of the Phoenicians who, on the orders of Pharaoh Necho,...
  - a) conquered Rome
  - b) reached America
  - c) sailed around Africa
  - d) discovered Australia
7. 💖 According to an anecdote, Thales of Miletus fell into a well because he was...
  - a) sleepwalking
  - b) counting his steps
  - c) chasing a cat
  - d) gazing at the stars
8. 🍃 Pliny the Elder believed that amber was...
  - a) fossilized tree resin
  - b) sea foam
  - c) tears of the gods
  - d) frozen honey
9. ☆ According to myth, Pandora opened a jar that released all the world's evils. Left at the bottom was...
  - a) wisdom
  - b) hope
  - c) love
  - d) gold